ASSIGNMENT

MODELING AND SIMULATION ~~TEST 1~~

Attempt **ALL** Questions ~~Time: 1.5 Hours~~

1. (a) Define performance in relation to computer systems.

   (b) Why is probability theory of interest to performance modeling of computer systems?

   (c) Define probability (either the experimental or axiomatic definition is sufficient).

   (d) A recent survey indicated that 80% of men aged 30 years old are married. Find the probability that 2 or 3 out of a random sample of 20 men of 30 years old are not married.

2. (a) Find the mean and variance of a Poisson distributed random variate.

   (b) Arrivals at a bank teller's cage are Poisson at the rate of 1.2 in a minute. Find the probability of:

   (i) No arrival in the next minute;

   (ii) 2 to 3 arrivals in the next 2 minutes.

3. (a) A compound has exponential time to failure with mean of 10,000 hours. The compound has already been in operation for 10,000 hours.

   (i) What is the probability that it will fail by 15,000 hours.

   (ii) If arrivals are Poisson, then show that inter-arrivals are exponential.

   (b) Assume that the length of a phone call in minutes is an exponential random variable X with parameter $\lambda = \frac{1}{10}$. If someone arrives at a phone booth just before you arrive, find the probability that you will have to wait:

   (i) less than 5 minutes;

   (ii) between 5 and 10 minutes.

4. A random variable X is said to be without memory, or memoryless, if

$$P(X \leq x + t | X > t) = P(X \leq x) \qquad x, t > 0 .$$

Show that if X is a nonnegative continuous random variable which is memoryless, then X must be an exponential variable.

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5. (a) For the following pdf, determine the Cumulative Distribution Function, $F(x)$.

$$f(x) = \begin{cases} \frac{1}{4}exp(4x), & -\infty < x \leq 0 \\ \frac{1}{4}, & 0 \leq x \leq \frac{15}{4} \\ 0, & \frac{15}{4} < x \end{cases}$$

(b) Assuming that X is a continuous random variable with the following pdf, determine α.

$$f(x) = \begin{cases} \frac{1}{\alpha}\exp(\alpha x), & -\infty < x \leq 0 \\ \frac{1}{\alpha}, & 0 \leq x \leq \frac{3}{\alpha} \\ 0, & \frac{3}{\alpha} < x \end{cases}$$

(c) If X has a mean of 5 and a variance of 4, Y has a mean of 4 and a variance of 1, and the correlation coefficient for X and Y is $\rho_{XY} = 0.3$, find:

(i) $E[-3X + 2Y + 10]$;

(ii) $Var[X + Y + 8]$.

6. (a) A production line manufactures 1000 Ω resistors that have 10% tolerance. Let X denote the resistance of a resistor. Assuming that X is a normal random variable with mean 1000 and variance 2500, find the probability that a resistor picked at random will be rejected?

(b) We repeat a simulation ten times and obtain the following values of one of the output variables.

1.051	6.438	2.646	0.805	1.505	0.546	2.281	2.822	0.414	1.307

Calculate a 95% confidence interval for the mean of these values.

(c) Which of these two statements is more accurate, and why?

I: "The probability that the mean is contained in your confidence interval is 95%".

II: "If you run more simulations and create many confidence intervals, then 95% of these intervals cover the true mean".

(d) For a given level of confidence α, how can you make the confidence interval shorter?

END.

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